

Project 4 Report

Currency Converter using Java

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Organization

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Objective

The objective of this project is to develop a Java-based Currency Converter application that enables users to convert an amount from one currency to another using predefined exchange rates. The application demonstrates the implementation of user input handling, arithmetic operations, conditional logic, and menu-driven programming while providing accurate conversion results.

Problem Statement

Develop a Java application that:

- * Accepts the source currency.
- * Accepts the target currency.
- * Accepts the amount to be converted.
- * Performs currency conversion.
- * Displays the converted amount.
- * Validates user inputs.

Software Requirements

- * IntelliJ IDEA
- * Java JDK 17+
- * Windows 10/11

Tools Used

- * Java
- * IntelliJ IDEA
- * Scanner Class

Project Description

The Currency Converter is a console-based Java application that allows users to convert an entered amount between supported currencies using predefined exchange rates. The application follows a menu-driven approach, validates user inputs, performs the required calculations, and displays the converted amount in a clean format. This project demonstrates practical implementation of Java fundamentals in solving a real-world problem.

Features

- * Currency selection
- * Amount input
- * Multiple currency support
- * Accurate conversion
- * Input validation
- * Menu-driven interface
- * Continuous execution until exit

Java Concepts Used

- * Variables
- * Scanner Class
- * Switch Case
- * Loops
- * Conditional Statements
- * Arithmetic Operations
- * Input Validation

Algorithm

1. Start the application.
2. Display available currencies.
3. Read source currency.

4. Read destination currency.
5. Accept the amount.
6. Validate the entered amount.
7. Perform currency conversion.
8. Display the converted value.
9. Repeat until the user chooses Exit.
10. End the application.

Expected Output

- * Currency selection menu
- * Amount input
- * Converted amount
- * Invalid input messages
- * Exit message

Learning Outcomes

- * Java console application development
- * User input handling
- * Currency conversion logic
- * Switch-case implementation
- * Arithmetic operations
- * Input validation
- * Problem-solving skills
- * Program structuring and debugging

Conclusion

The Currency Converter project successfully demonstrates the application of core Java programming concepts through the development of a practical console-based utility. By implementing user input handling, currency conversion logic, validation, and a menu-driven interface, the project provides a solid understanding of Java fundamentals while meeting the objectives of the Decode Labs Java Programming Internship Project 4.